MIL-S-60502A(MU)
18 June 1970
SUPERSEDING
MIL-S-60502(MU)
5 August 1966

MILITARY SPECIFICATION

STOP PREAMPLIFIER: 10540715

1. SCOPE

1.1 This specification covers a solid state wideband pulse amplifier which is a modular component of the Range Finder, Laser:AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacturing and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-I-46058	Insulating Compound, Electrical for Coating Printed Circuit Assemblies

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

B10540715	Stop Preamplifier
C10540705	Housing (Mounting Pads and Hole Positions)
C10540747	Connector
D10540831	Electrical Schematic

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PACKAGING DATA SHEET

10540715 Packaging of Stop Preamplifier

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The stop preamplifier hereinafter referred to as amplifier shall be manufactured in accordance with Drawing B10540715 and drawings pertaining thereto.

3.3 General specification.- The amplifier shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Soldering electrical connections
e. Part identification and marking
f. Workmanship

3.3.1 Electronic workmanship.- In addition to meet the workmanship requirements of MIL-F-13926, the amplifier shall meet the electronic workmanship requirements of 3.3.1.1 through 3.3.1.12.

3.3.1.1 Etched circuit board.- The conductor pattern on the etched board shall be free from blisters, wrinkles, cracks, holes, dents, scratches and any other defects that would affect serviceability. The pattern shall be free of loosely adhering feather edges of metal. In no case shall a finishes line width be more than 0.010 inch narrower than that specified in the printed circuit master. In no case shall any conductor line width be less than 0.020 inch.

3.3.1.2 Spacing of parts.- Unless otherwise specified on the applicable drawing, all parts shall be mounted with the component in contact with the board at least at one point. The component shall not be raised more than 0.031 inch from the board at any point on its contacting surface.

3.3.1.3 Symbols and markings.- Unless physically impossible, all parts shall be mounted on the board in such a manner that any reference symbols on the board are not obscured, and that the proper identification markings of each part are visible.

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3.3.1.4 Transistors.- Transistors shall be mounted perpendicular to the circuit board with spacers between the transistors and circuit board.

3.3.1.5 Bend radius.- The minimum bend radius for all formed leads shall be one lead diameter; the maximum shall be two diameters or 0.062 inch, whichever is smaller.

3.3.1.6 Component leads.- Leads of transistors mounted with a spacer directly onto a printed circuit board, shall have 1/16 inch of lead from the transistor base to any soldered joint. Component leads shall be performed to suit mounting holes. There shall be visible evidence of leads passing completely through the circuit board. The leads shall not protrude more than 3/64 inch, including solder, beyond the side of the circuit board opposite the component.

3.3.1.7 Plated-through connections.- All plated-through connections shall be filled with solder in order to eliminate dependence on the plating itself for circuit continuity.

3.3.1.8 Heat sensitive components.- The use of heat shunts or cooling devices shall be mandatory when soldering components that are subject to damage by heat.

3.3.1.9 Cleaning.- After soldering operations have been completed, the assembly shall be thoroughly cleaned and rinsed, the cleaning fluid shall have a gentle flow, at no time exceeding a pressure of 5 psi. The temperature of the cleaning solution shall not exceed 125°F. After cleaning, there shall be no residual flux, surface contamination, or detrimental staining on the assembly. Cleaned assemblies shall not be allowed to become contaiminated after cleaning, nor shall markings become illegible.

3.3.1.10 Drying.- The assembly may be dried by exposure to room temperature without the aid of auxiliary heating or air circulating equipment. If other than room temperature drying method is used, the assembly shall be dried in a circulating air oven, the temperature of which shall not exceed 165°F. When compressed air or nitrogen is used for drying parts, the pressure applied to the parts shall not exceed 5 psi. The air or nitrogen shall be filtered to meet the following requirements.

a. Permissible dust particle size not to exceed 6 microns
b. Permissible moisture content limited to 2.12 grains per cubic foot (dew point 32°F maximum)
c. Dust count shall not exceed 1000 particles per cubic foot

3.3.1.11 Coating.

3.3.1.11.1 Masking.- Prior to coating with insulating compound in accordance with MIL-I-46058, the test point TP-1 shall be masked on top and sides to protect from coating.

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3.3.1.11.2 Coating, insulating compound.- The printed circuit board assembly with electrical components installed, shall be coated with an insulating compound. The insulating compound shall be Insulating Compound, Electrical, Type PUR, Polyurethane resin conforming to MIL-I-46058. The exterior wire harness shall not be coated with the resin.

3.3.1.11.3 Cured compound.- The cured compound shall represent a smooth continuous surface and shall pass from the mounted components to the circuit board with no break at their junction. Except for area masked, all surfaces shall be covered with a coating of 0.001 to 0.006 inch thickness. The cured compound shall meet the following requirements.

(a) Compound shall be free of visible dust particles or other foreign matter.
(b) It shall be smooth and uniform and have no globules or bare spots.
(c) It shall be substantially free of bubbles.
(d) There shall be no blisters or cracks.
(e) It shall be transparent and shall not affect marking legibility.
(f) It shall be capable of being removed in local areas for replacement of components by a soldering iron.
(g) Applied to a test board of 1/16 inch thickness, the compound shall not rupture or separate from the board when the board is bent into an arc of twelve inches radius.

3.3.1.12 Wire termination.- When soldering wires to terminals or solder points a small gap of 1/32 inch minimum to 1/16 inch maximum shall be left between the wire insulation and the solder build-up. Burned, frayed, or otherwise damaged wire insulation shall not be acceptable. Stripping of insulation shall be accomplished without disarranging, cutting, or otherwise damaging any strand of the conductor. Heat shall not be used to remove outer jacket from shielded wire. Only mechanical methods or removal are permitted.

EXCEPTION: The hot wire stripping method may be used where the wire or cable primary insulation and jacket are made of Teflon (TFE) or equal.

3.4 Vibration.- The amplifier shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the amplifier shall show no evidence of physical damage and shall meet the requirements of 3.7 through 3.9.

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3.5 Shock.- The amplifier shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the amplifier shall show no evidence of physical damage, and shall meet the requirements of 3.7 through 3.9.

3.6 Storage temperatures.- The amplifier shall withstand thermal stabilization at temperatures of -80° Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the amplifier shall show no evidence of physical damage, and shall meet the requirements of 3.7 through 3.9.

3.7 Performance.- The amplifier shall meet the requirements of 3.7.1 to 3.7.6, while thermally stabilized at temperatures of -65°F, +155°F, and at a temperature between +70° and +80°F, when operated with a supply voltage of 5.5 ± 0.1 VDC.

3.7.1 Pulse response.- The amplifier shall generate a negative pulse 1.5V minimum amplitude, 100 ± 10 nsec wide (Figure 1) to a 240 ohm load connected across Pins E and B of P1 when a 40 ± 10 nsec wide pulse (Figure 2), which is more negative than the input threshold amplitude of 3.7.2, is applied to the Input P2.

3.7.2 Input threshold.- The input threshold of the amplifier shall be controlled for the time initiated by the application to Pin C of P1 of a 1.25 ± 0.25V amplitude, 0.5 ± 0.1 usec pulse (Figure 3). For a period of 1 usec to 4.5 ± 0.25 usec from initiation of the control, the amplifier shall give the response of 3.7.1 to negative pulses which are equal to or greater than the following limits:

a. 40 ± 2 mv at temperature between 70°F and 80°F.
b. 50 ± 2 mv at temperature of +155°F.
c. 60 ± 2 mv at temperature of -65°F.

The threshold at all other temperatures shall not exceed 60 mv. From the time 4.5 ± 0.25 usec to 7.5 ± 0.25 usec the threshold shall decrease exponentially to a final limit of 2 ± 1 mv.

3.7.3 Propagation delay.- The elapsed time between the application of the input pulse to P2 and the emergence of an output pulse at Pin E of P2 shall not exceed 60 nsec (see Figure 4).

3.7.4 Stability.- With the input open circuited the amplifier shall attain a steady state condition of no output transitions exceeding 0.25V within 50 msec after the application of supply voltage.

3.7.5 Spurious outputs.- Outputs at Pin E, other than those specified in 3.7.1 and 3.7.2, shall not exceed 0.25V in amplitude for all times following one usec after application of the initiation pulse to Pin C.

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3.7.6 Input current.- The current required by the amplifier shall not exceed 30 ma.

3.8 Electrical continuity.- The electrical wiring of the amplifier shall conform to the Electrical Schematic Drawing D10540831.

3.9 Interface requirements.- The amplifier shall interface with:

a. Mounting pads and holes located in Housing C10540705.
b. Connector C10540747 (13J4) of Telescope Assembly.
c. Connector, Amphenol, Type 5116-037550 (12P2) of tube, photomultiplier.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 assemblies manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for

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materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS		REQUIREMENT	TEST PROCEDURE
CRITICAL:	NONE		
MAJOR:	AQL 1.0% DEFECTIVE		
101.	Electronic workmanship	3.3.1	Visual
102.	Etched circuit board	3.3.1.1	SME; Visual
103.	Spacing of parts	3.3.1.2	SME; Visual
104.	Symbols and markings	3.3.1.3	Visual
105.	Transistors	3.3.1.4	Visual
106.	Bend radius	3.3.1.5	SME; Visual
107.	Component leads	3.3.1.6	SME; Visual
108.	Plated-through connections	3.3.1.7	Visual
109.	Heat sensitive components	3.3.1.8	Visual
110.	Masking	3.3.1.11.1	Visual
111.	Cured compound	3.3.1.11.3	Visual-Tactile
112.	Wire termination	3.3.1.12	Visual
MINOR:	NONE		

NOTE: The above characteristics shall be inspected at in-process intermediate inspection stations.

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TABLE II - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
113.	Performance	3.7	4.6.4
114.	Pulse response	3.7.1	4.6.4.1
115.	Input threshold	3.7.2	4.6.4.1
116.	Propagation delay	3.7.3	4.6.4.2
117.	Stability	3.7.4	4.6.4.3
118.	Spurious outputs	3.7.5	4.6.4.4
119.	Input current	3.7.6	4.6.4.5
120.	Electrical continuity	3.8	4.6.5
121.	Interface requirements	3.9	4.6.6

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristics specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract product, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Cleaning	3.3.1.9
Drying	3.3.1.10
Coating, insulating compound	3.3.1.11.2

4.4 Special sampling.

4.4.1 General.- Three amplifiers shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual

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NOTE: The tests in Table III shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three amplifiers shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table IV and shall then meet the requirements and tests in Table II.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Vibration (+70°F and +80°F)	3.4	4.6.3
305.	Shock (+70°F and +80°F)	3.5	4.6.2
306.	Storage temperatures (-80°F & +160°F)	3.6	4.6.1.1
307.	Performance (-65'F, +155°F)	3.7 through 3.7.6	4.6.4 through 4.6.4.5

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does

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not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2 the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

	NOMENCLATURE	DESCRIPTION
1.	Signal source	A pulse generator with two output channels capable of producing a positive pulse and a delayed negative pulse. The pulses shall be as described in Figures 2 and 3. The delay between the outputs of the two channels shall be adjustable from 0 to 10 microsec and the output of the negative channel shall be adjustable from 0 to 100 mv.
2.	Oscilloscope	Rise time 6 nanoseconds, input impedance of 10 meg ohms or greater. Sweep time of 100 nanoseconds per centimeter, sensitivity of 5 mv or less per centimeter, equipped with a camera.
3.	Ammeter, D.C.	Accuracy ± 2% of reading, to measure input current as specified in 3.8.2.

4.6 Test methods and procedures.

4.6.1 Environmental.- The amplifiers shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the amplifier shall be subjected to the tests specified in 4.6.4 through 4.6.4.5.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.6.

4.6.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.

4.6.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The amplifier shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.4. The vibration test shall then

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be performed in accordance with Method 514 Procedure 1, Part 1, of MIL-STD-810.

4.6.4 Performance.- Conduct the tests in 4.6.4.1 to 4.6.4.5 at all temperatures specified in 3.7 with 5.5 ± 0.1 VDC applied between Pins A and B of Connector P1, (Pin A is +5.5 VDC).

4.6.4.1 Pulse response and input threshold.- Using pulse generator (Item 1 of 4.5.3) apply the pulses specified in 3.7.1 and 3.7.2 at a repetition rate which allows for convenient monitoring of the output on an oscilloscope. Adjust the separation of these input pulses to 1 usec and adjust the amplitude of the input pulse to P2 to within the limits of the input threshold (3.7.2). The resultant output pulse shall conform to 3.7.1. Increase the separation of the input pulses toward 7.5 usec. As the separation of pulses exceeds 4.5 usec, the input pulse amplitude required to produce the output of 3.7.1 shall conform to 3.7.2.

4.6.4.2 Propagation delay.- Synchronize the oscilloscope with the leading edge of the pulse applied to P2 and monitor the output produced at Pin E of Connector P1. The time between application of an input pulse to P2 and the emergence of an output pulse at P1 shall conform to 3.7.3.

4.6.4.3 Stability.- Remove all input connections from P2. Monitor Pin E of P1 with an oscilloscope for a period of one minute. The amplifier shall conform to 3.7.4.

4.6.4.4 Spurious outputs.- Remove all input connections from P2 and monitor Pin E of P1 with an oscilloscope. Apply the initiation pulse of Figure 3 to Pin C of P1. Except for an output pulse occurring within 1 usec of the application of the pulse to Pin C, there shall be no outputs from the amplifier as specified in 3.7.5.

4.6.4.5 Input current.- Using the DC ammeter (Item 3 of 4.5.3), measure the current supplied to Pin A of P1 of the amplifier. This current shall not exceed the requirements of 3.7.6.

4.6.5 Electrical continuity.- The continuity check of the amplifier circuit shall be made using standard measuring equipment to determine compliance with the requirements of 3.8.

4.6.6 Interface requirements.- Perform this test using two selected test housings. One housing selected for the maximum limit of each interface with a tolerance of 10% of the total limit, and one selected to the minimum limit of interface beyond maximum or minimum limits. The amplifier shall be capable of being properly positioned and secured to both test housings, while mated with the Nixie Power Supply Module 10540716, utilizing the Mounting Clip 10540036. The housing assembly interfaces shall be as depicted on Drawing C10540705 and drawings pertaining thereto. The Connectors Drawing C10540747 shall mate with their respective receptacles

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to a depth of at least .09 inch and show continuity with all pertinent electrical connections.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10540715. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - Signals from the photomultiplier tube of the Range Finder, AN/GVS-3 are amplified and controlled by the stop preamplifier for use by the time interval counter's stop channel, and target counting circuits.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of applicable levels of preservation, packaging and packing.
c. Applicable packaging data sheet (see 5.1).
d. Provisions for first artile testing.

6.3 Definitions.-

F = Fahrenheit
Kohm = 1000 ohms
V = volts
VDC = volts direct current
mv = millivolts
msec = milliseconds
usec = microseconds
nsec = nanoseconds
min.= minimum
max.= maximum

Custodian:

Army - MU

Preparing activity:

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OUT PUT- PULSE

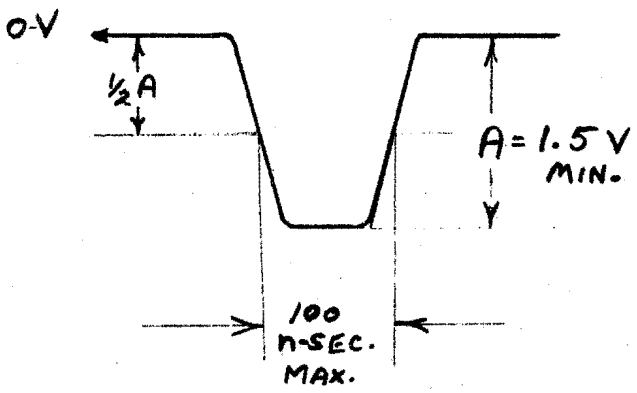


FIG. 1

INPUT-PULSE

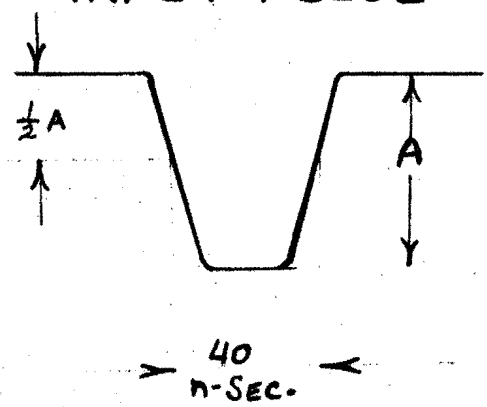


FIG. 2

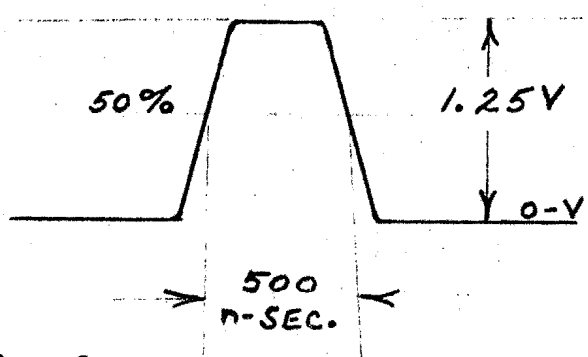


FIG. 3

INITIATION-PULSE

DELAY-PULSE

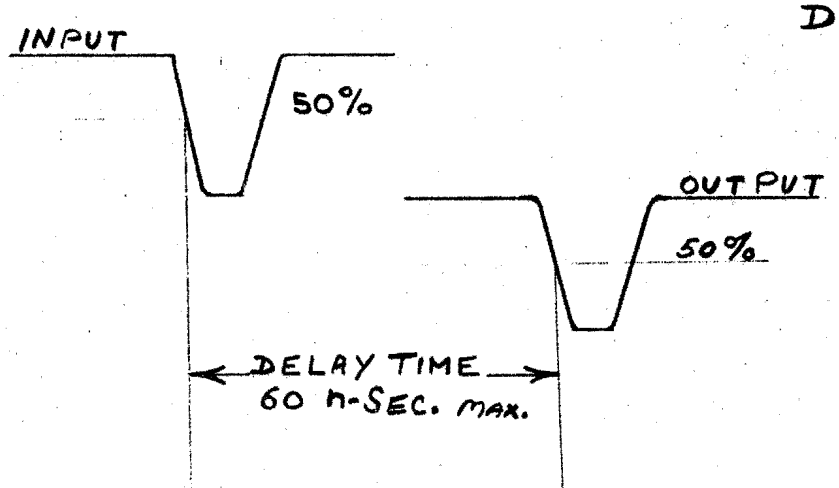


FIG. 4

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